

Nandam Josh Nihanth

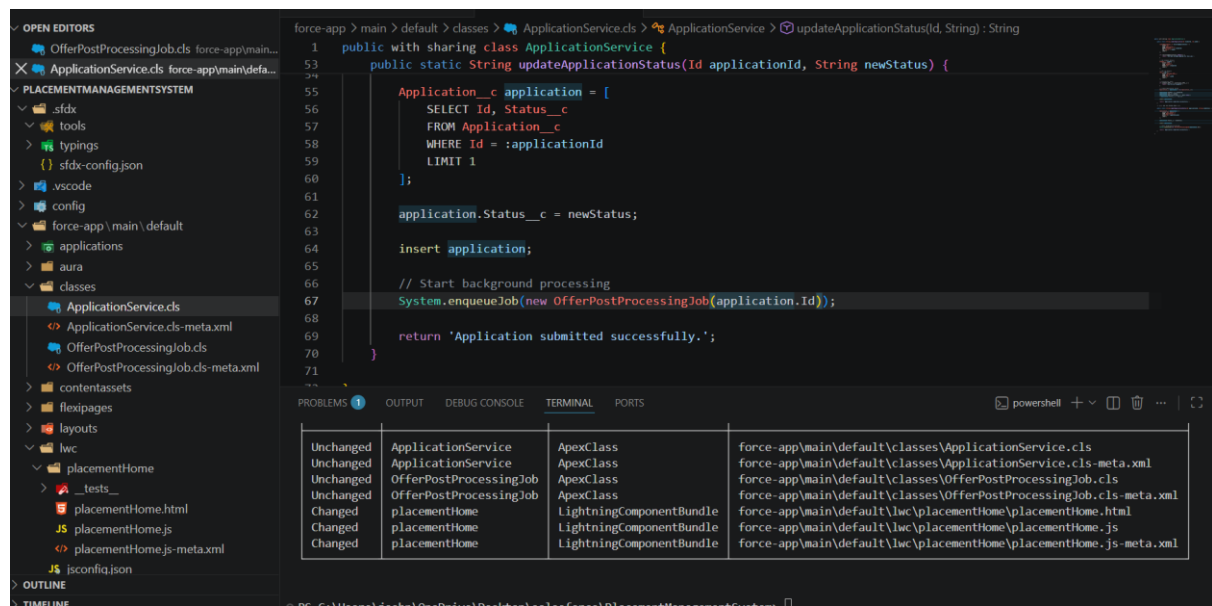
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CSE-C

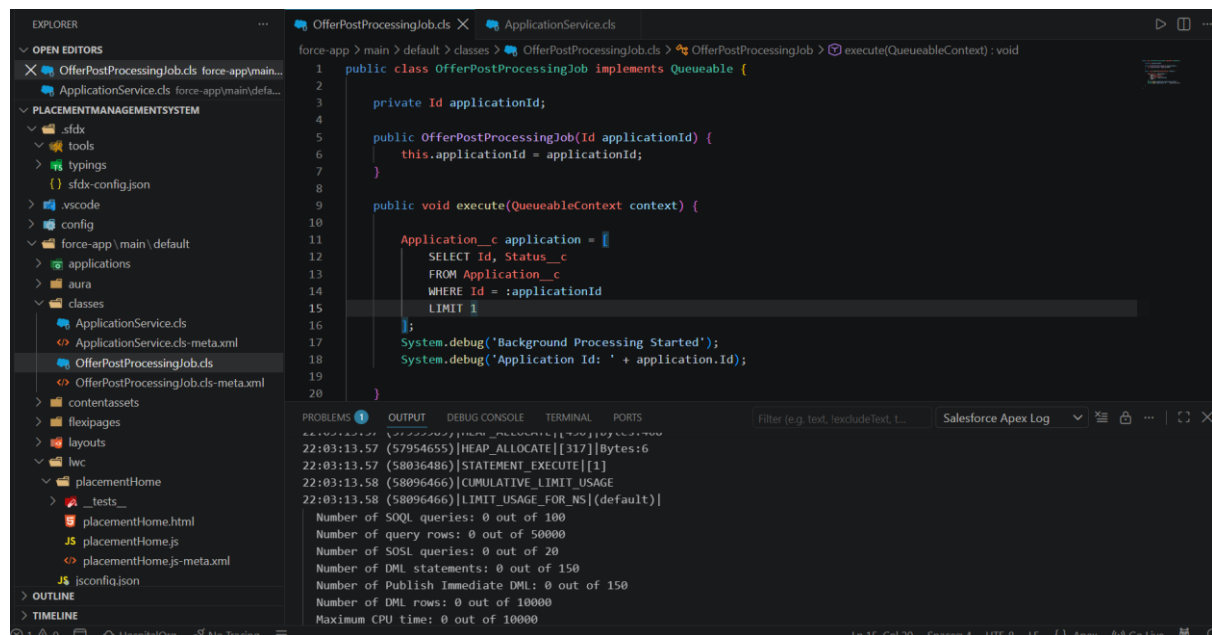
DAY-08

SALESFORCE TRAINING PROGRAM

In this day, I learned about Asynchronous Apex processing in Salesforce and how it helps in executing operations in the background without affecting the user's current transaction.



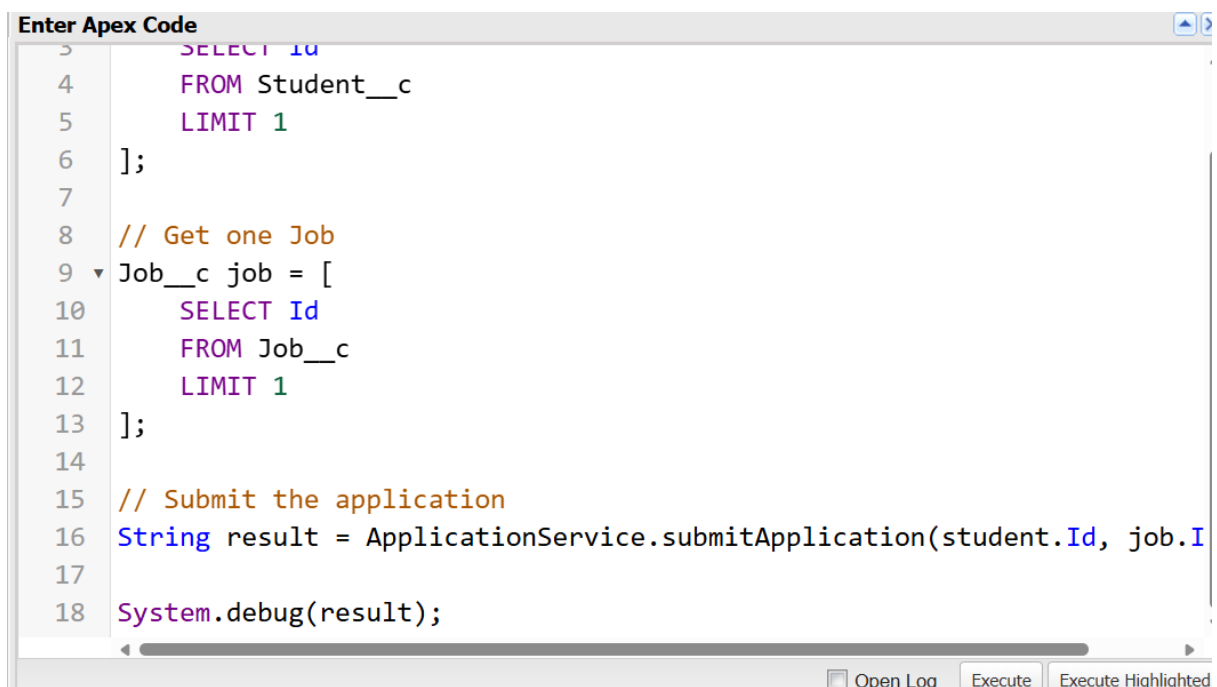
In real-time Salesforce applications, some operations may take more time or involve processing a large amount of data. Performing such operations synchronously can slow down the user experience and may hit Salesforce governor limits.



To overcome this challenge, Salesforce provides asynchronous processing mechanisms such as:

- Future Methods
- Queueable Apex
- Batch Apex
- Scheduled Apex

As part of this day's practical implementation, I extended the Placement Management System by implementing background processing using Queueable Apex, Queueable Chaining, Batch Apex, and Scheduled Apex.



```
1  SELECT Id
2  FROM Student__c
3  LIMIT 1
4  ];
5
6  // Get one Job
7  Job__c job = [
8      SELECT Id
9      FROM Job__c
10     LIMIT 1
11 ];
12
13 // Submit the application
14 String result = ApplicationService.submitApplication(student.Id, job.Id);
15
16 System.debug(result);
```

Open Log Execute Execute Highlighted

Synchronous processing means that the operation is executed immediately in the same transaction.

Execution Log		
Timestamp	Event	Details
22:34:29:080	USER_DEBUG	[32] DEBUG Batch Apex Completed Successfully.

Student submits application



Application record created



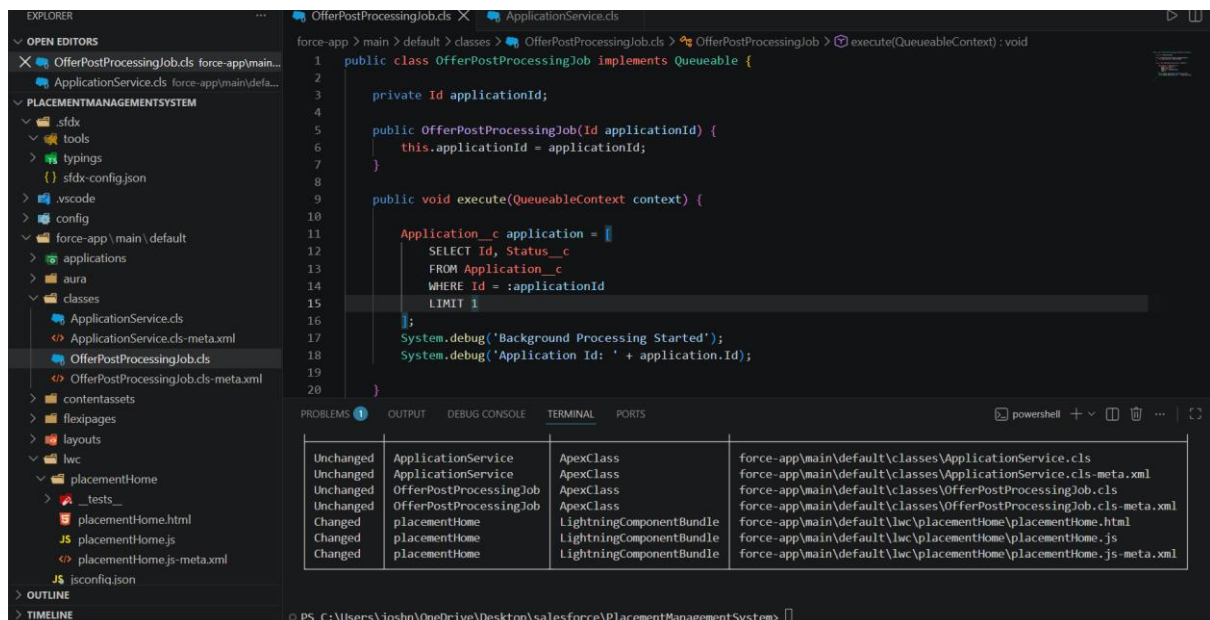
Response returned to user

Simple execution flow.

Immediate result.

Time-consuming operations can slow down the user experience.

Governor limits can be reached easily when processing large data.



Asynchronous processing allows Salesforce to execute operations separately in the background.

The user transaction completes first, and Salesforce processes the additional work later.

Execution Log		
Timestamp	Event	Details
22:30:44:197	USER_DEBUG	[17][DEBUG]You have already applied for this job.

Enter Apex Code

```

1  ▼ Student__c student = [
2      SELECT Id, Name
3      FROM Student__c
4      LIMIT 1
5  ];
6
7  ▼ Job__c job = [
8      SELECT Id, Name
9      FROM Job__c
10     LIMIT 1
11  ];
12
13  System.debug('Student Id: ' + student.Id);
14  System.debug('Job Id: ' + job.Id);

```

☐ On

Time ▼ Status Read

Queueable Apex is used when we want to execute a task in the background after the main transaction is completed.

Execution Log		
Timestamp	Event	Details
22:05:45:029	USER_DEBUG	[13][DEBUG]Student Id: a07dM00001ZoWj4QAF
22:05:45:029	USER_DEBUG	[14][DEBUG]Job Id: a08dM00000ltPKVQA2

In the Placement Management System, after creating an application record, additional processing is moved to a Queueable job.

applicationId stores the record Id that needs to be processed.

The constructor receives the Application Id.

The execute() method runs when Salesforce processes the Queueable job.

The application record is queried again to get the latest data.

Execute Anonymous Error

Line: 46, Column: 1
System.DmlException: Insert failed. First exception on row 0; first error:
CANNOT_INSERT_UPDATE_ACTIVATE_ENTITY, ApplicationTrigger: execution of BeforeInsert caused by:
line 4, column 28: Method does not exist or incorrect signature: void
validateApplications(List<Application__c>) from the type ApplicationService: []

OK

File • Edit • Debug • Test • Workspace • Help • < >

Log executeAnonymous @7/8/2026, 10:05:45 pm ApplicationTrigger.apxt * Log executeAnonymous @7/8/2026, 10:22:39 pm

Execution Log

Timestamp	Event	Details
22:22:39:013	USER_DEBUG	[18][DEBUG]Background Processing Started
22:22:39:013	USER_DEBUG	[19][DEBUG]Application Id: a09dM00000VUiyTQAT

☐ This Frame
☐ Executable
☒ Debug Only
☐ Filter
Click here to filter the log

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Read	Size
NANDAM JOSH NIHANATH	Unknown	QueueableHandler	7/8/2026, 10:22:39 pm	Success		3.89 KB

EXPLORER

OfferPostProcessingJob.cls ApplicationNotificationJob.cls ApplicationService.cls

force-app > main > default > classes > ApplicationNotificationJob.cls > ...

```
9 public void execute(QueueableContext context) {
10
11     Application__c application = [
12         SELECT Id, Status__c
13         FROM Application__c
14         WHERE Id = :applicationId
15         LIMIT 1
16     ];
17
18     System.debug('Background Processing Started');
19     System.debug('Application Id: ' + application.Id);
20
21     // Chain the next Queueable Job
22     System.enqueueJob(new ApplicationNotificationJob(application.Id));
23 }
24
```

PLACEMENTMANAGEMENTSYSTEM

- .sfdx
- tools
- sqlMetadata
- testresults
- typings
- sfdx-config.json
- .vscode
- config
- force-app\main\default
- applications
- aura
- classes

ApplicationNotificationJob.cls

ApplicationNotificationJob.cls-meta.xml

ApplicationService.cls

ApplicationService.cls-meta.xml

OfferPostProcessingJob.cls

OfferPostProcessingJob.cls-meta.xml

contentassets

flexipages

layouts

lwc

OUTLINE

TIMELINE

node + - [] [] ...

Changed ApplicationService ApexClass force-app\main\default\classes\ApplicationService.cls-meta.xml

Changed OfferPostProcessingJob ApexClass force-app\main\default\classes\OfferPostProcessingJob.cls

Changed placementHome LightningComponentBundle force-app\main\default\lwc\placementHome\placementHome.html

Changed placementHome LightningComponentBundle force-app\main\default\lwc\placementHome\placementHome.js

Changed placementHome LightningComponentBundle force-app\main\default\lwc\placementHome\placementHome.js-meta.xml

PS C:\Users\joshn\OneDrive\Desktop\salesforce\PlacementManagementSystem> sf project deploy start

Warning: @salesforce/cli update available from 2.143.6 to 2.145.6.

Log executeAnonymous @7/8/2026, 10:05:45 pm ApplicationTrigger.apxt * Log executeAnonymous @7/8/2026, 10:22:39 pm

Execution Log

Timestamp	Event	Details
22:22:39:013	USER_DEBUG	[18] DEBUG Background Processing Started
22:22:39:013	USER_DEBUG	[19] DEBUG Application Id: a09dM00000VUiyTQAT

☐ This Frame ☐ Executable ☒ Debug Only ☐ Filter Click here to filter the log

Logs Tests Checkpoints Query Editor View State Progress Problems

User	Application	Operation	Time	Status	Rea
NANDAM JOSH NIHANTH	Unknown	Batch Apex	7/8/2026, 10:34:29 pm	Success	
NANDAM JOSH NIHANTH	Unknown	SerialBatchApexRangeChunkHandl...	7/8/2026, 10:34:29 pm	Success	Unre
NANDAM JOSH NIHANTH	Unknown	SerialBatchApexRangeChunkHandl...	7/8/2026, 10:34:29 pm	Update failed. First exception on r...	Unre
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/exec...	7/8/2026, 10:34:28 pm	Success	Unre
NANDAM JOSH NIHANTH	Unknown	Batch Apex	7/8/2026, 10:34:28 pm	Success	Unre
NANDAM JOSH NIHANTH	Unknown	/services/data/v67.0/tooling/exec...	7/8/2026, 10:32:32 pm	Success	Unre

☐ Filter Click here to filter the log list

GITHUB :[LINK](#)